May 10, 2026

Editorial Office
Nature Electronics
Springer Nature

Dear Editors,

Submission: Hardware-Aware Training and Precision-Retention Frontiers for Low-Precision Analog Vision

Article Type: Article

Editorial Summary

We present an open-source simulation framework for organic optoelectronic compute-in-memory inference with modern vision transformers. Our central contribution, Ensemble HAT, resamples device-to-device mismatch masks every epoch. On Tiny-ViT-5M, it raises fresh-instance accuracy from chance level (10.00%) to 86.16% (mean ± 0.19%), enabling robust deployment without per-device calibration.

Post-fix severe-NL retraining recovers to the ∼80–82% band, falsifying the previously reported ∼30% ceiling and localizing it to a resolved software artifact. A residual gap of roughly 15 pp to the digital baseline remains, whose origin—physical limit vs algorithmic frontier—awaits higher-order surrogates or measured device calibration.

Novelty Claims

- We introduce the first open-source framework for organic optoelectronic CIM with vision transformers, providing PyTorch-native device-profile substitution and hybrid analog/digital deployment.
- We identify hardware-instance overfitting in organic CIM inference and propose Ensemble HAT, a training-time mismatch-mask resampling fix that restores generalization.
- Post-fix severe-NL retraining recovers to the ∼80–82% band, falsifying the previously reported ∼30% ceiling, with a residual gap whose closure awaits higher-order surrogates or measured device data.

Audience

Nature Electronics readers will find a reusable methodology for evaluating organic device assumptions on vision tasks and a rigorous negative-result anchor that delineates the limits of first-order behavioral simulation.

Related Submissions

This work is not under consideration elsewhere.

Reviewer Information

Suggested and excluded reviewers are entered through the submission system.

Contact Information

Li Songqiao / 2622507532@qq.com

Sincerely,

Songqiao Li